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AGRI NOVA MARKET ENTRY ANALYSIS

An Econometric Study of Revenue Determinants and Product Launch Strategy

Name: Abdullah Ibrahim

Course: Economics And Business Analytics

Assignment: Individual Report (40%)

Date: 21 April 2026

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1. Management Summary

This study looks at the primary economic variables that affect sales income for CropSense and FreshConnect, two products that AgriNova has suggested. Regression analysis is used to determine which product should be introduced first based on data-driven insights and to identify what generates revenue for each product. With extremely high adjusted R2 values, the results show that both models successfully explain revenue. But the factors that affect each product are very different.

Prices for agricultural commodities and advertising have a major impact on CropSense's revenue. Another important factor is price, therefore changes in price have a significant impact on demand. This suggests that pricing choices must be handled carefully. Further scenario analysis reveals that raising digital advertising and changing prices might increase income by about 6.6%.

On the other hand, consumer-related variables like household income, digital advertising, and population density have a greater impact on FreshConnect. One important discovery is that FreshConnect's demand is not highly sensitive to price fluctuations, making price tweaks easier to implement without significantly affecting demand.

All things considered, FreshConnect seems to be the better choice for the first launch. It has less short-term risk, depends on steady demand drivers, and provides more steady and predictable revenue.

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2. Introduction (Industry Overview)

Over the past ten years, urbanization, technological advancement, and changing customer preferences have all contributed to the fast and steady growth of both the fresh food delivery business and the agritech sector. The need for effective food production and distribution systems has grown dramatically as populations continue to congregate in metropolitan areas. New business models that close the gap between producers and customers have also been made possible by advances in digital technologies.

Agritech solutions, such data-driven decision-making platforms, precision farming tools, and Internet of Things (IoT)-based monitoring systems, are becoming more and more popular among farmers. By maximizing resource utilization, cutting waste, and raising crop yields, these technologies increase productivity. For instance, real-time monitoring of soil conditions and weather patterns using smart sensors enables farmers to make better decisions about fertilization and irrigation. In areas where resource limitations and environmental issues are major considerations, these technologies are especially crucial.

In response to shifting customer habits, the fresh food delivery sector has expanded concurrently. Convenience, quality, and openness in food supply are becoming more important to urban customers, particularly those with higher incomes. The idea of "farm-to-table" has grown in popularity as customers want locally sourced, organic, and fresh goods delivered straight to their homes. Businesses may now communicate with customers more easily, handle logistics, and guarantee on-time delivery thanks to digital platforms.

Southeast Asian governments have been instrumental in helping both industries. They have promoted the adoption of contemporary agricultural technologies and the growth of digital marketplaces through governmental initiatives, subsidies, and infrastructural expenditures. These initiatives have made it easier for businesses like AgriNova to launch cutting-edge goods and services.

These two marketplaces have quite different demand structures from an economic standpoint. Farmers are the main users of agritech products like CropSense since they frequently have limited resources and are consequently more susceptible to price fluctuations. Customers of fresh food delivery services, such as FreshConnect, on the other hand, are generally less price-sensitive because they value quality and convenience over cost.

Making strategic decisions requires an understanding of these distinctions. According to economic theory, demand typically declines as prices rise; nevertheless, each market has a different level of responsiveness, or price elasticity (Mankiw, 2021). By measuring the influence of important variables like price, income, and advertising on revenue, regression analysis offers a methodical way to investigate these correlations. This helps businesses make well-informed choices about product introductions, marketing expenditures, and pricing tactics.

3. Report Objectives

This report's primary objective is to determine and assess the economic aspects influencing CropSense and FreshConnect's sales revenue.

More precisely, the paper seeks to:

- Determine which important factors have a major influence on revenue.
- Estimate the price elasticity of demand for both products
- Analyze the effects of digital and traditional advertising
- Evaluate the regression models' performance using R^2 and adjusted R^2
- Conduct scenario analysis to explore potential revenue changes
- Recommend the most suitable product for initial launch

4. Data Analysis

4.1 Exploratory Analysis and Bivariate Relationships

Simple linear regressions were used in the first phase of the analysis to look at each independent variable's unique relationship to revenue. This method aids in separating each factor's impact and offers a basic understanding of the variables that have the most impact on CropSense and FreshConnect's revenue.

With an R2 value of almost 0.75, the results show that the agricultural commodity price index is the most significant explanatory variable for CropSense. This indicates that changes in the larger agricultural market account for about 75% of the variation in CropSense revenue. This substantial correlation emphasizes how dependent CropSense is on outside economic factors, especially those that impact farmers' earnings and purchasing power.

1 The explanatory power of advertising variables, both digital and traditional, is moderate. This suggests that while marketing initiatives do help generate income, their influence is not as strong as that of the commodity price index. Price and income, on the other hand, seem to have comparatively smaller individual effects when examined separately, indicating that their impact can become more noticeable when taken into account in conjunction with other factors.

FreshConnect has a very distinct pattern. The most important factor is household income, which has an R2 value at 0.98. This shows that income and revenue have an almost perfect linear connection, indicating FreshConnect's heavy reliance on customer purchasing power. Furthermore, digital advertising has significant explanatory power ($R^2 = 0.91$), highlighting the significance of online marketing in connecting with urban customers.

However, price alone has little explanatory power for FreshConnect, indicating that price fluctuations have little effect on demand. Demand patterns are influenced by urban concentration, as evidenced by the comparatively lesser but still discernible impact of population density.

These results unequivocally show that FreshConnect is largely driven by consumer-related factors like income and digital engagement, whereas CropSense is impacted by external market conditions.

4.2 Graphical Analysis

The associations between important variables were visually examined using graphical representations to supplement the regression analysis. These graphs offer intuitive insights on trends, patterns, and relationship strength.

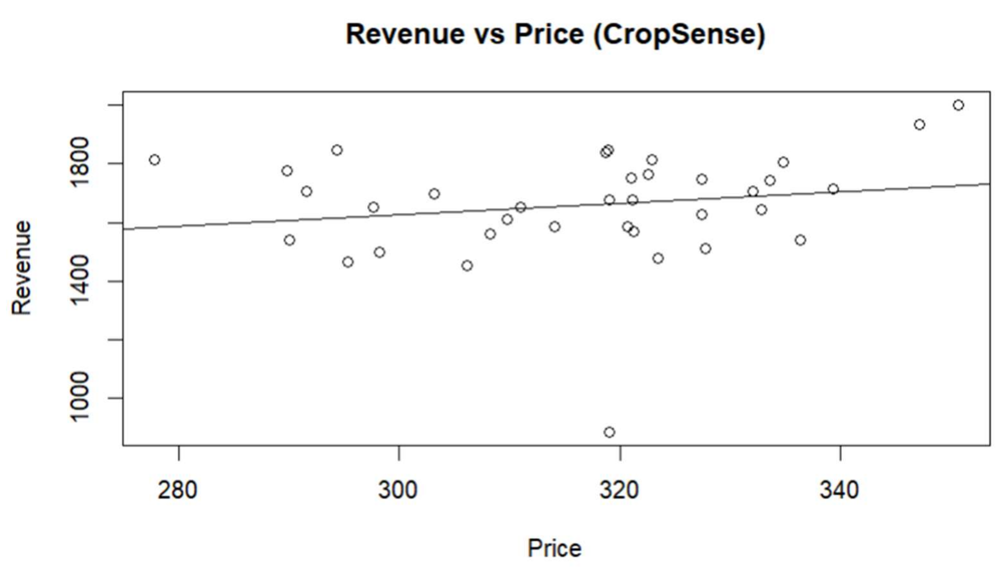

Figure 1: Revenue vs Price (CropSense)

CropSense revenue and price have a comparatively weak and somewhat positive association, according to the scatter plot. Although price and demand should have a negative relationship according to economic theory, the observed pattern reveals that this relationship is not very strong in this dataset.

The data points are widely distributed, indicating revenue fluctuation that price cannot adequately account for. The fitted trendline just slightly slopes upward, suggesting that price increases are not always linked to appreciable drops in revenue. This could be because the direct impact of pricing is overshadowed by other prominent factors like market circumstances and commodity prices.

The existence of outliers is another crucial finding, especially at specific price points when revenue greatly deviates from the overall trend. These anomalies indicate that pricing decisions should not be made in a vacuum and further emphasize the relationship's intricacy.

Overall, the graph confirms the previous conclusion that other factors are more important in determining CropSense revenue than price.

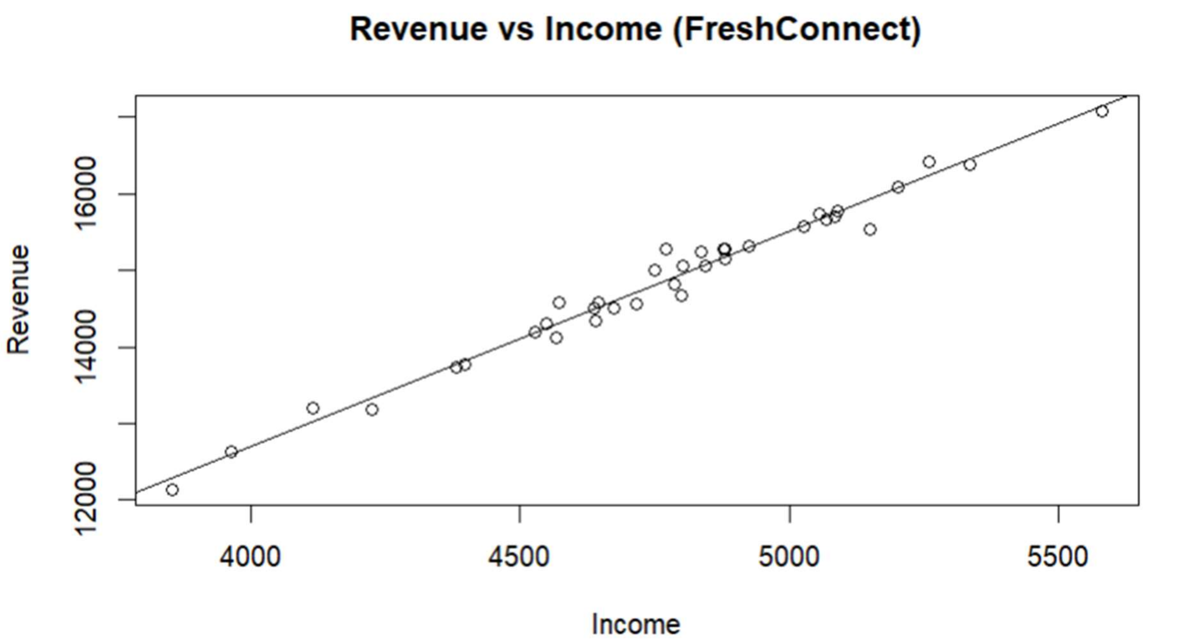

Figure 2: Revenue vs Income (FreshConnect)

There is a clear and robust positive linear link between household income and FreshConnect revenue in the scatter plot. A high level of correlation and consistency is indicated by the data points' close clustering around the fitted regression line.

Revenue increases nearly proportionately to income, indicating that FreshConnect functions like a typical good. Convenience-based services like fresh food delivery are more likely to be purchased by consumers with higher income levels. Income is a reliable predictor of revenue, as evidenced by the tight clustering of points around the trendline, which also shows minimal variability.

The regression results are strongly supported by this graphical data, demonstrating that the most important factor influencing FreshConnect demand is income. It also implies that focusing on affluent urban customers would be a successful tactic for increasing profits.

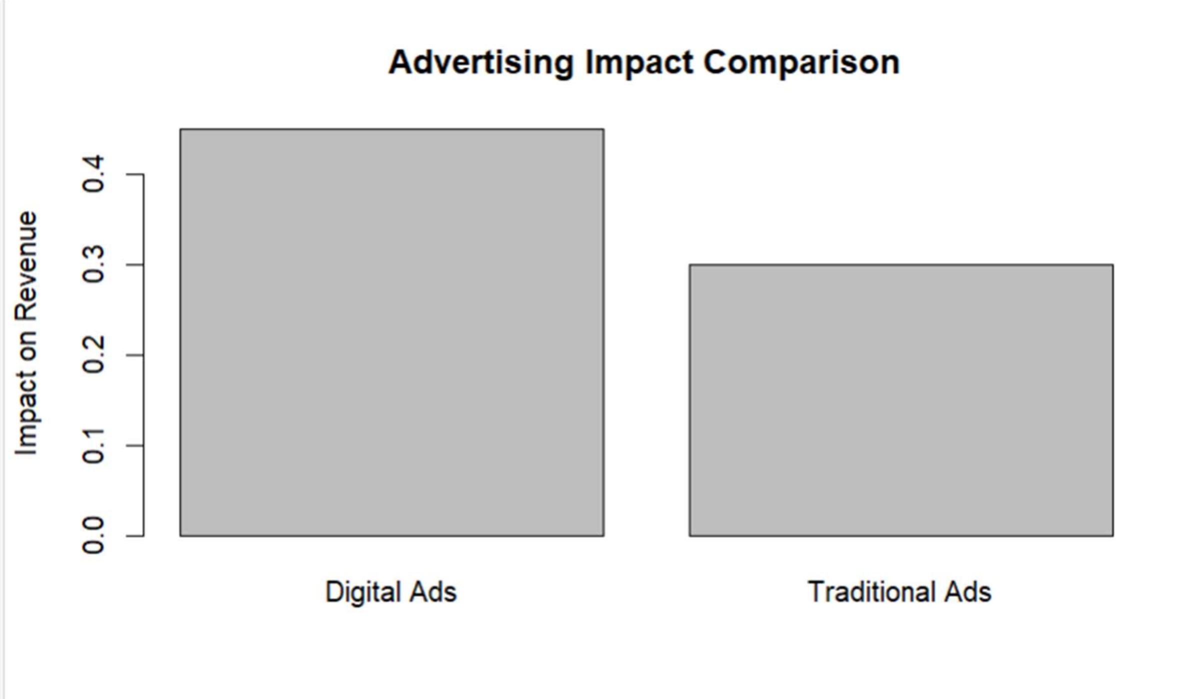

Figure 3: Advertising Impact Comparison

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There is a noticeable difference in the efficacy of digital and traditional advertising, according to the bar graph that compares their effects. When compared to traditional advertising, digital advertising has a greater influence on revenue, suggesting that online platforms are more effective at reaching and persuading consumers.

FreshConnect, which operates in an urban, digitally linked setting where customers are more receptive to internet marketing campaigns, will find this study very pertinent. Even if traditional advertising still makes a good contribution, its comparatively smaller impact indicates that resources should be more carefully directed toward digital platforms.

It is clear from the visual contrast between the two bars that digital advertising is a major factor in revenue development and ought to be given top priority in marketing plans.

4.3 Multiple Regression Model Results

The outcomes of a multiple regression framework become more thorough and representative of actual circumstances when all variables are included at the same time.

The model's adjusted R2 for CropSense is roughly 0.99, meaning that the factors included account for almost all of the variation in income. According to the law of demand, price and revenue have a statistically significant negative connection. But compared to the commodity price index, which continues to be the most important determinant, its impact is comparatively less.

The positive and statistically significant coefficients of both digital and traditional advertising attest to their contribution to increasing income. It's interesting to note that family income is not statistically significant in this model, indicating that agricultural conditions have a greater influence on CropSense demand than consumer income.

With an improved R2 near 0.999, the model performs even better for FreshConnect. The most important predictor is household income, which is followed by digital advertising. The concept that demand is comparatively insensitive to price fluctuations is reinforced by the negative but negligible impact of price. Additionally, population density has a beneficial impact, suggesting that the product works especially well in cities.

4.4 Price Elasticity Analysis

Price elasticity gives an indication of how responsive demand is to price fluctuations. CropSense has a moderately elastic demand, with an estimated elasticity of about -0.39. This implies that both demand and revenue can be significantly impacted by price fluctuations.

FreshConnect, on the other hand, has a very low elasticity of about -0.02. This suggests extremely inelastic demand, where price adjustments have no impact on customer behavior. This discrepancy implies that pricing choices are more important for CropSense than for FreshConnect, which has important strategic ramifications.

4.5 Statistical Significance and Confidence Intervals

The statistical analysis confirms that most variables are significant at the 5% level, indicating that the observed relationships are unlikely to be due to random chance. These conclusions are further supported by confidence intervals, which show that important factors like price and advertising have dependable and consistent effects on revenue.

For FreshConnect, income and population density show strong and stable impacts, while for CropSense, the commodity price index remains the most robust predictor. These findings bolster the models' legitimacy and validate their application in decision-making.

4.6 Economic Logic Validation

The actual results are in good agreement with accepted economic theory. While the positive effect of advertising emphasizes the role of marketing in influencing customer behavior, the negative relationship between price and demand validates the law of demand.

Given its heavy reliance on income, FreshConnect appears to be a typical good, with demand rising in tandem with consumer income. In the meantime, the impact of external market conditions on agricultural demand is reflected in CropSense's dependence on commodity prices.

The validity of the analysis is strengthened by the agreement between theory and empirical findings.

4.7 Scenario Analysis

To assess how changes in important variables impact CropSense's income outcomes, scenario analysis was carried out.

In the first scenario, a substantial income boost of roughly 6.64% was achieved by combining lower prices with more digital advertising. This illustrates how well-coordinated pricing and marketing initiatives work.

In the second case, cutting back on advertising costs only slightly increased sales by 0.51%, underscoring the significance of continuing to make significant marketing investments.

These results highlight the importance of pricing and digital advertising as levers for increasing revenue and enhancing performance.

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5. Discussion

It is evident from the analysis that CropSense and FreshConnect function rather differently.

CropSense is more reliant on outside variables, especially the cost of agricultural commodities. Because of its modest price elasticity, pricing choices are important and have a big influence on demand. Higher gains are possible as a result, but there is also more risk.

In contrast, FreshConnect is more reliable and consistent. Income, advertising, and population density—all of which are comparatively constant—are what propel its demand. Pricing adjustments have little impact on demand due to the low price elasticity, which facilitates management.

FreshConnect is the safer option from a business standpoint. It provides scalability and consistent growth, particularly in metropolitan markets. CropSense has the potential to be profitable, but it needs more careful planning and is more susceptible to outside influences.

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6. Recommendations

A number of strategic recommendations for AgriNova can be made in light of the investigation. First, as FreshConnect exhibits steady and predictable demand driven by income and digital engagement, it should be given priority for the initial market launch. Its low price elasticity lowers risk and permits more adaptable pricing tactics.

Second, the two products should be priced differently. Because of its modest price sensitivity and reliance on external market conditions, especially changes in commodity prices, CropSense necessitates careful pricing judgments. FreshConnect, on the other hand, permits more pricing freedom because demand is less vulnerable to price fluctuations.

Third, the company's marketing approach should be centered on digital advertising. According to the data, digital channels have a greater effect on revenue than traditional advertising, particularly for FreshConnect's target urban consumers. CropSense can still use conventional advertising in agricultural markets, though.

Lastly, it's important to establish market segmentation precisely. CropSense should concentrate on agricultural areas, whereas FreshConnect should target affluent urban consumers. It is advised that CropSense be introduced gradually following more data analysis and market assessment.

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7. Conclusion

Regression analysis was used in this study to investigate the major factors influencing CropSense and FreshConnect's income. Despite the great explanatory power of both models, the results show that their demand structures differ significantly.

CropSense shows low price elasticity and is mostly impacted by external market conditions, especially the agricultural commodity price index. This increases risk and complexity by making it more susceptible to changes in the market and price decisions. FreshConnect, on the other hand, exhibits a steady and predictable demand pattern that is influenced by consumer-related variables including family income, digital advertising, and population density.

These conclusions are further supported by the graphical analysis, especially the robust positive correlation between income and revenue for FreshConnect. Because of its incredibly low price elasticity, demand appears to be mostly unaffected by price fluctuations, enabling more strategic flexibility.

Overall, because of its lower risk, more predictability, and scalability in urban settings, FreshConnect is determined to be the better solution for initial market entry. The results highlight how crucial data-driven decision-making is for maximizing corporate goals and guaranteeing long-term growth.

8. References

- Basic Microeconomic Theory (Demand and Elasticity)
- Regression Analysis Methods
- Course Lecture Notes

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9. Annexures

- R^2 Comparison Table
- Regression Outputs
- Confidence Intervals
- Scenario Analysis Results